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Course Code: CSA1402

Compiler Design Lab Programs

33.School management wants to validate DOB of all students. Write a LEX program to implement it.

Code:

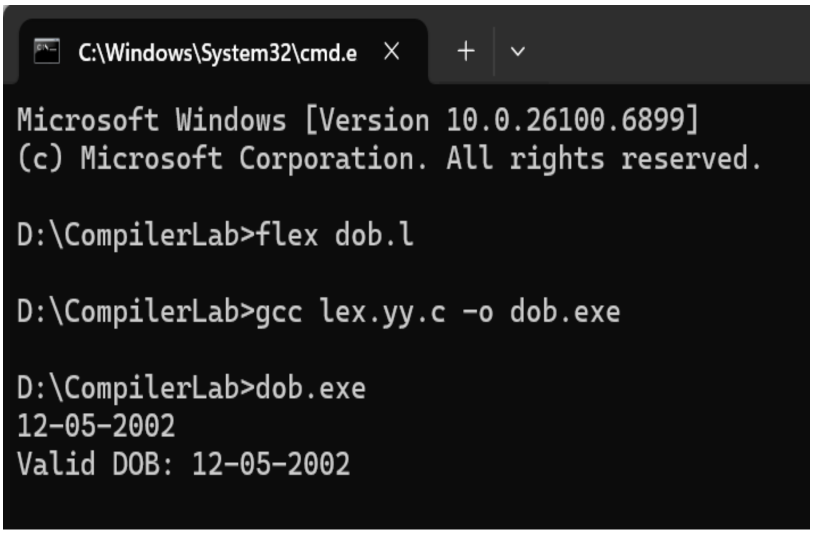
```
%{
#include <stdio.h>
%}
%%
(0[1-9]|[12][0-9]|3[01])[-/.](0[1-9]|1[0-2])[-/.](19[0-9]{2}|20[0-9]{2}) {
    printf("Valid DOB: %s\n", yytext);
}
[^\n]+ { printf("Invalid DOB: %s\n", yytext); }
\n ;
%%
int main(){ yylex(); return 0; }
int yywrap(){ return 1; }
```

Output:

12-05-2002

Valid DOB: 12-05-2002

Output Screenshot:



34.Write a LEX program to check whether the given input is digit or not.

Code:

```
%{
#include <stdio.h>
%}
%%
[0-9]   { printf("It is a Digit: %s\n", yytext); }
.|\n    { printf("Not a Digit: %s\n", yytext); }
%%
int main(){ yylex(); return 0; }
int yywrap(){ return 1; }
```

Output:

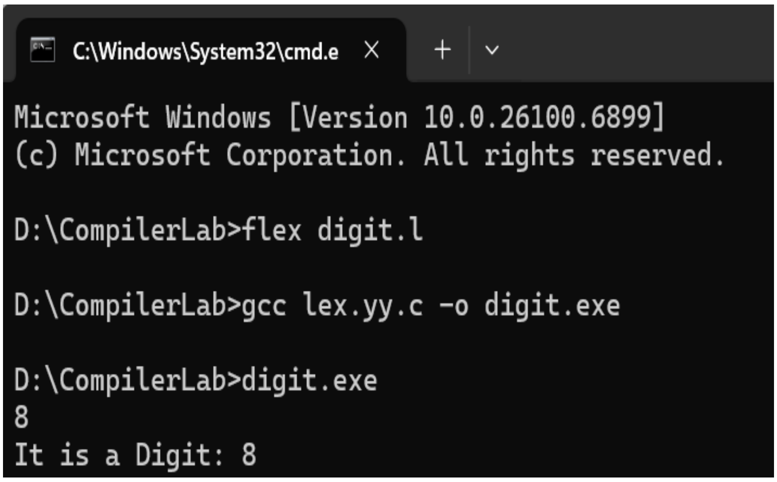
8

a

It is a Digit: 8

Not a Digit: a

Output Screenshot:



35.A School student was asked to do basic mathematical operations. Implement a LEX program to implement the same.

Code:

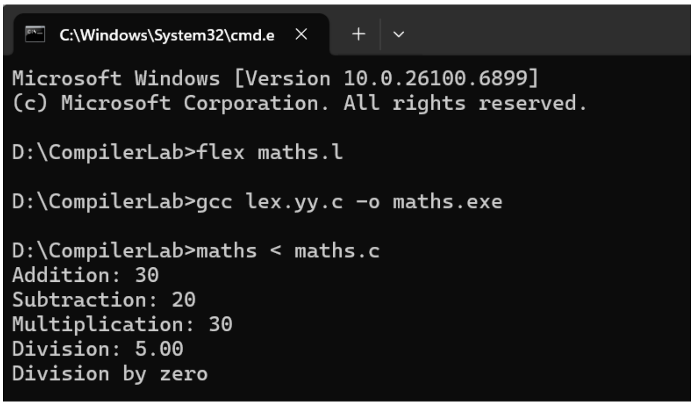
```
%{
#include <stdio.h>
#include <stdlib.h>
```

```
%}
NUMBER  [0-9]+
%%
{NUMBER}"+"{NUMBER} { int a,b; sscanf(yytext,"%d+%d",&a,&b); printf("Addition: %d\n", a+b); }
{NUMBER}"-"{NUMBER} { int a,b; sscanf(yytext,"%d-%d",&a,&b); printf("Subtraction: %d\n", a-b); }
{NUMBER}"*"{NUMBER} { int a,b; sscanf(yytext,"%d*%d",&a,&b); printf("Multiplication: %d\n", a*b); }
{NUMBER}"/"{NUMBER} { int a,b; sscanf(yytext,"%d/%d",&a,&b); if(b!=0) printf("Division: %.2f\n", (float)a/b); else printf("Division by zero\n"); }
[ \t\n]+  ;
.        ;
%%
int main(){ yylex(); return 0; }
int yywrap(){ return 1; }
```

Output:

```
10+20
25-5
6*5
20/4
20/0
Addition: 30
Subtraction: 20
Multiplication: 30
Division: 5.00
Division by zero
```

Output Screenshot:



36.Write a LEX program to accept string starting with vowel.

Code:

```
%{
#include <stdio.h>
%}
%%
[AEIOUaeiou][a-zA-Z]*    { printf("Valid: %s\n", yytext); }
[a-zA-Z]+                { printf("Invalid: %s\n", yytext); }
.|\n                     ;
%%
int main(){ yylex(); return 0; }
int yywrap(){ return 1; }
```

Output:

apple

tree

Orange

Valid: apple

Invalid: tree

Valid: Orange

Output Screenshot: